Assignment4.2

2026-03-01

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
library("modelsummary")
```

```
## Warning: package 'modelsummary' was built under R version 4.5.2
```

```r
library("marginaleffects")
```

```
## Warning: package 'marginaleffects' was built under R version 4.5.2
```

```r
library(ggplot2)
install.packages("readstata13", repos = "http://cran.us.r-project.org")
```

```
##
## The downloaded binary packages are in
##  /var/folders/s0/j16mvfvs79l6ql5nkpgn7y000000gn/T//RtmpQoiym6/downloaded_packages
```

```r
library(broom)



df <- readstata13::read.dta13("https://raw.githubusercontent.com/franvillamil/AQM2/refs/heads/master/da

summary(df$country)
```

```
##    Length     Class      Mode
##       101 character character
```

```r
unique(df$region)
```

```
## [1] "Asia"     "Europe"   "Americas" "Africa"
```

```r
summary(df$income)
```

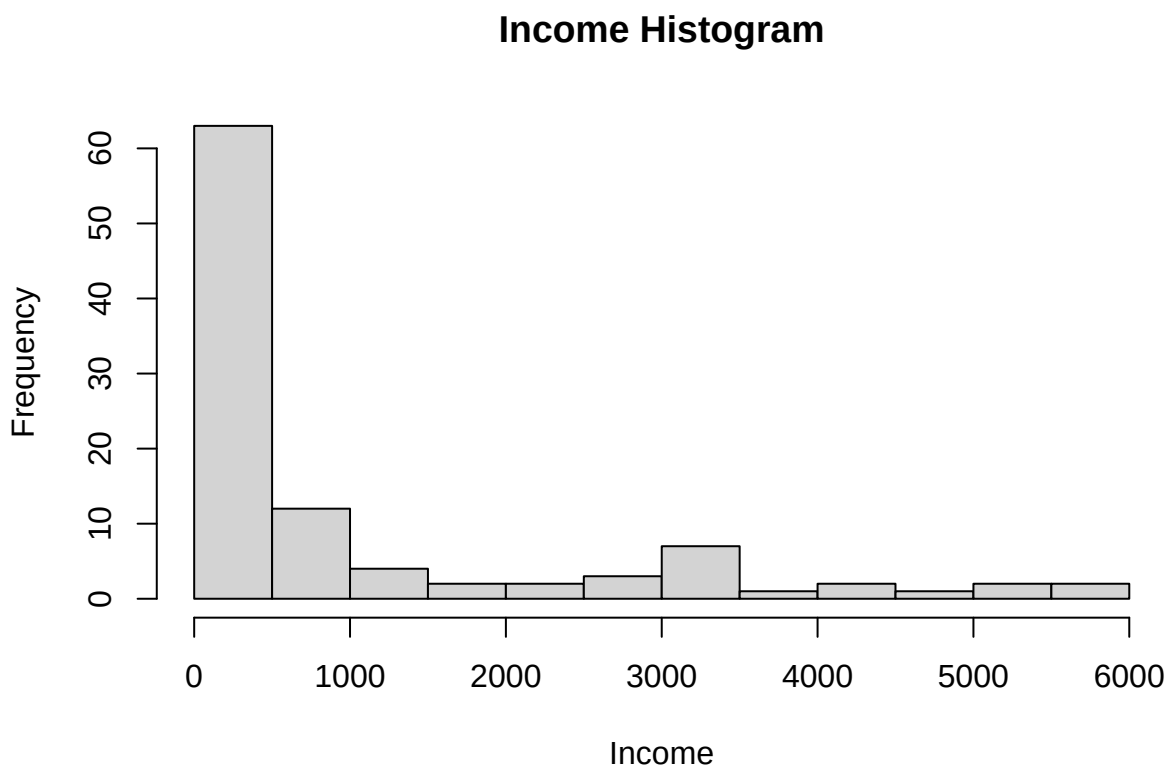
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      50     130     334    1022    1191    5596
```

```r
summary(df$infant)
```

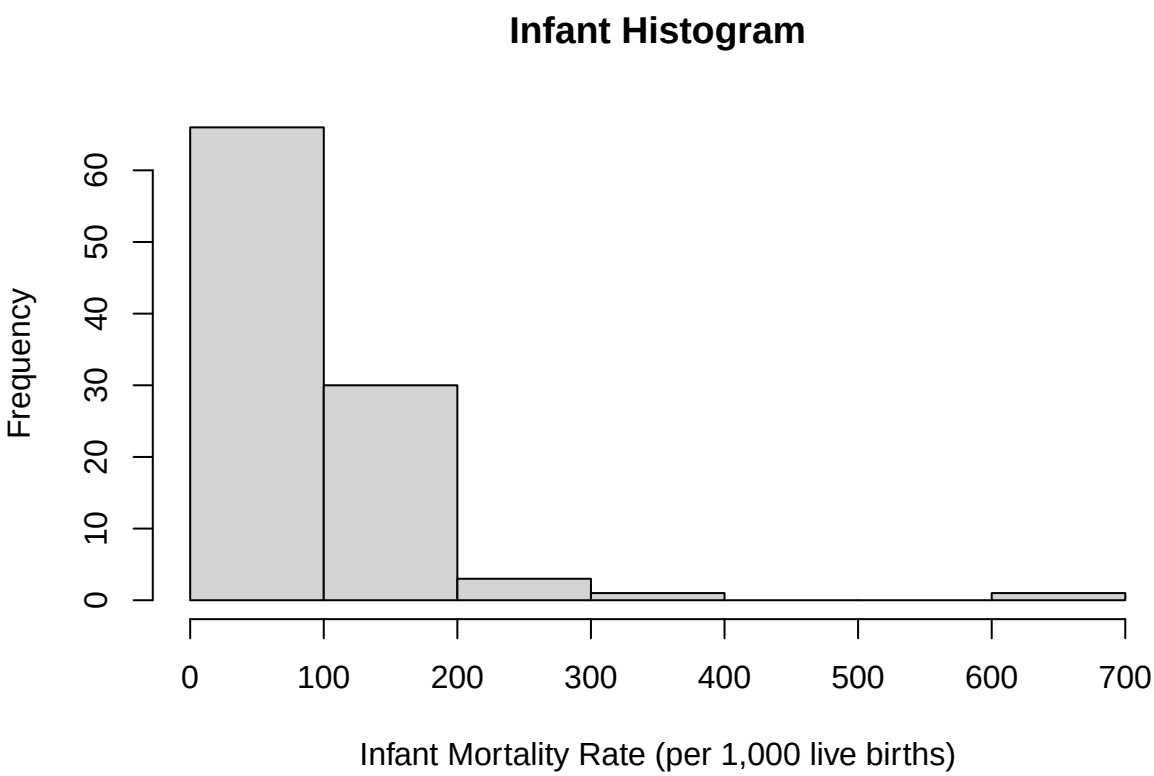
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    9.60   26.20   60.60   89.05  129.40  650.00
```

```r
# 101 countries in the set

hist(df$income, xlab = "Income", main = "Income Histogram")
```

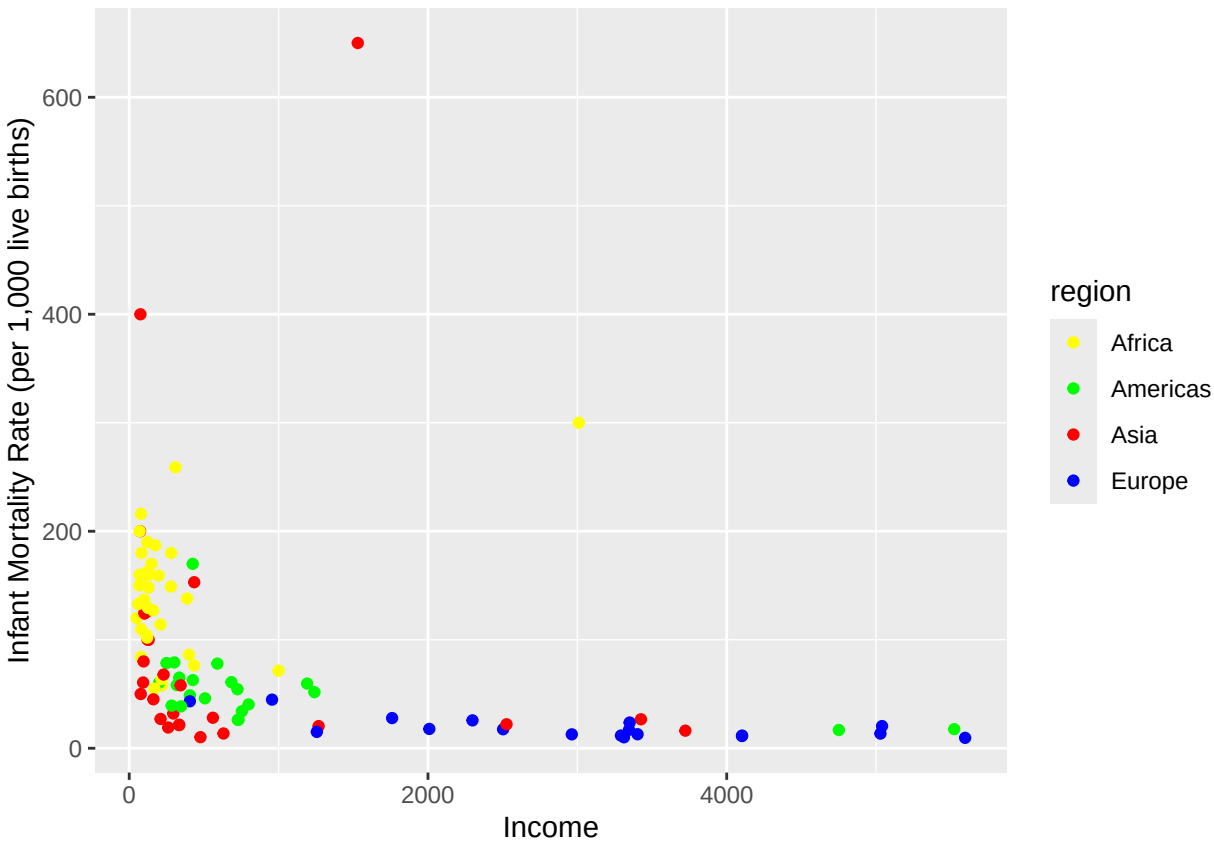



```r
hist(df$infant, xlab = "Infant Mortality Rate (per 1,000 live births)", main = "Infant Histogram")
```



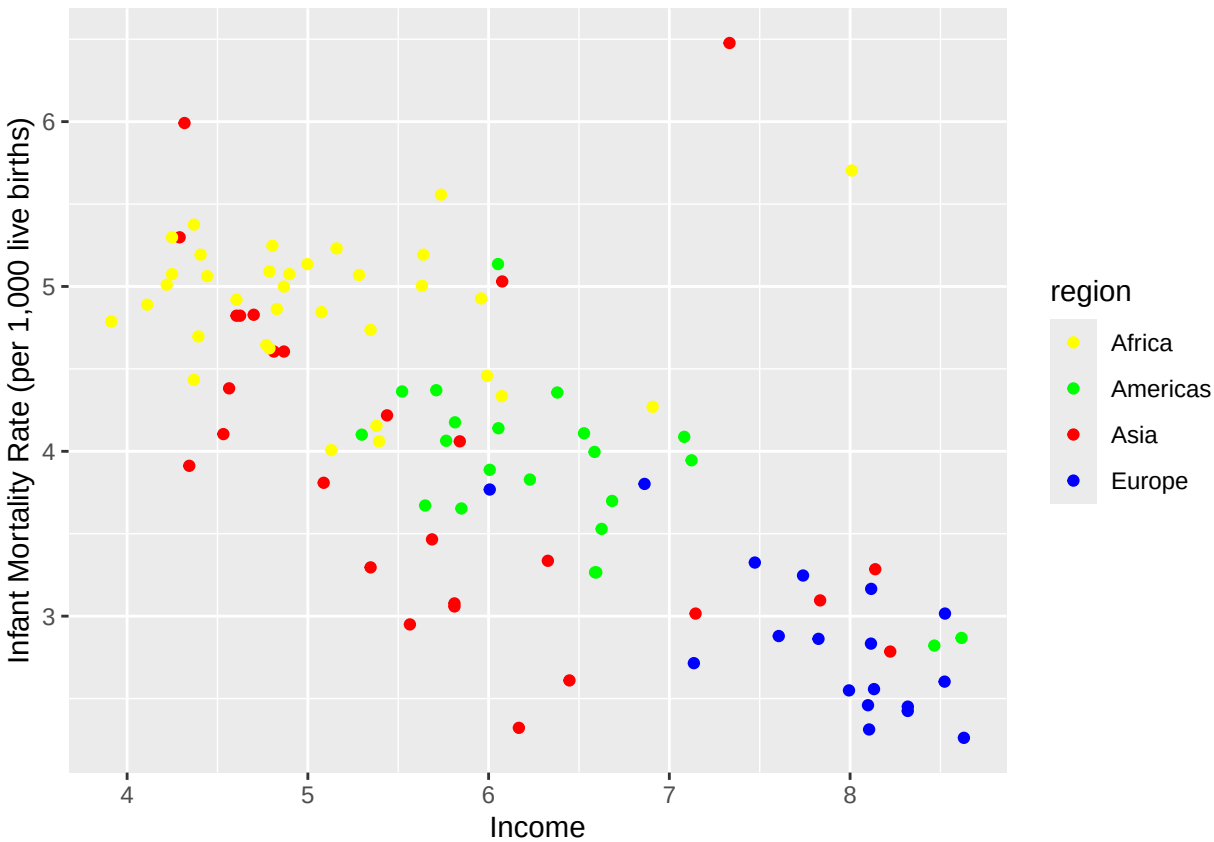
Both variables "Infant" and "Income" are right skewed, we know this because the graphs peaks are more towards the left and because the means are bigger than the medians.

```
ggplot(df, aes(x=income, y = infant, color = region))+
  geom_point()+
  labs(
    x = "Income",
    y = "Infant Mortality Rate (per 1,000 live births)"
  ) +
  scale_color_manual(values = c("Asia" = "red", "Europe" = "blue",
                                "Americas" = "green", "Africa" = "yellow"))
```



Africa generally (with a couple outliers) has low income and relatively high infant mortality rates. Europe has countries with various incomes but mostly all low infant mortality rates. The Americas are clustered at low incomes (with some outliers) and low infant mortality rates. Lastly, Asia has more variety and has many outliers in terms of high income and for high infant mortality rates too.

```
ggplot(df, aes(x=log(income), y = log(infant), color = region))+
  geom_point()+
  labs(
    x = "Income",
    y = "Infant Mortality Rate (per 1,000 live births)"
  ) +
  scale_color_manual(values = c("Asia" = "red", "Europe" = "blue",
                                "Americas" = "green", "Africa" = "yellow"))
```



This graph seems more linear than the last and is spread moreso that it is easier to see the relationships for each region. Yet the outliers are still obvious and we cannot be sure yet that the relationship(s) is(are) linear.

```
m1 = lm(infant ~ income, data = df)
summary(m1)
```

```
##
## Call:
## lm(formula = infant ~ income, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -90.25 -45.84 -23.47  22.65 571.57
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 110.421086  10.538807  10.478  < 2e-16 ***
## income       -0.020907   0.005999  -3.485 0.000735 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 86.13 on 99 degrees of freedom
## Multiple R-squared:  0.1093, Adjusted R-squared:  0.1003
## F-statistic: 12.14 on 1 and 99 DF,  p-value: 0.0007354
```

```
m2 = lm(log(infant) ~ log(income), data = df)
summary(m2)
```

```
##
```

```
## Call:
## lm(formula = log(infant) ~ log(income), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.66694 -0.42779 -0.02649  0.30441  3.08415
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.14582    0.31654  22.575   <2e-16 ***
## log(income) -0.51179    0.05122  -9.992   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6867 on 99 degrees of freedom
## Multiple R-squared:  0.5021, Adjusted R-squared:  0.4971
## F-statistic: 99.84 on 1 and 99 DF,  p-value: < 2.2e-16
```

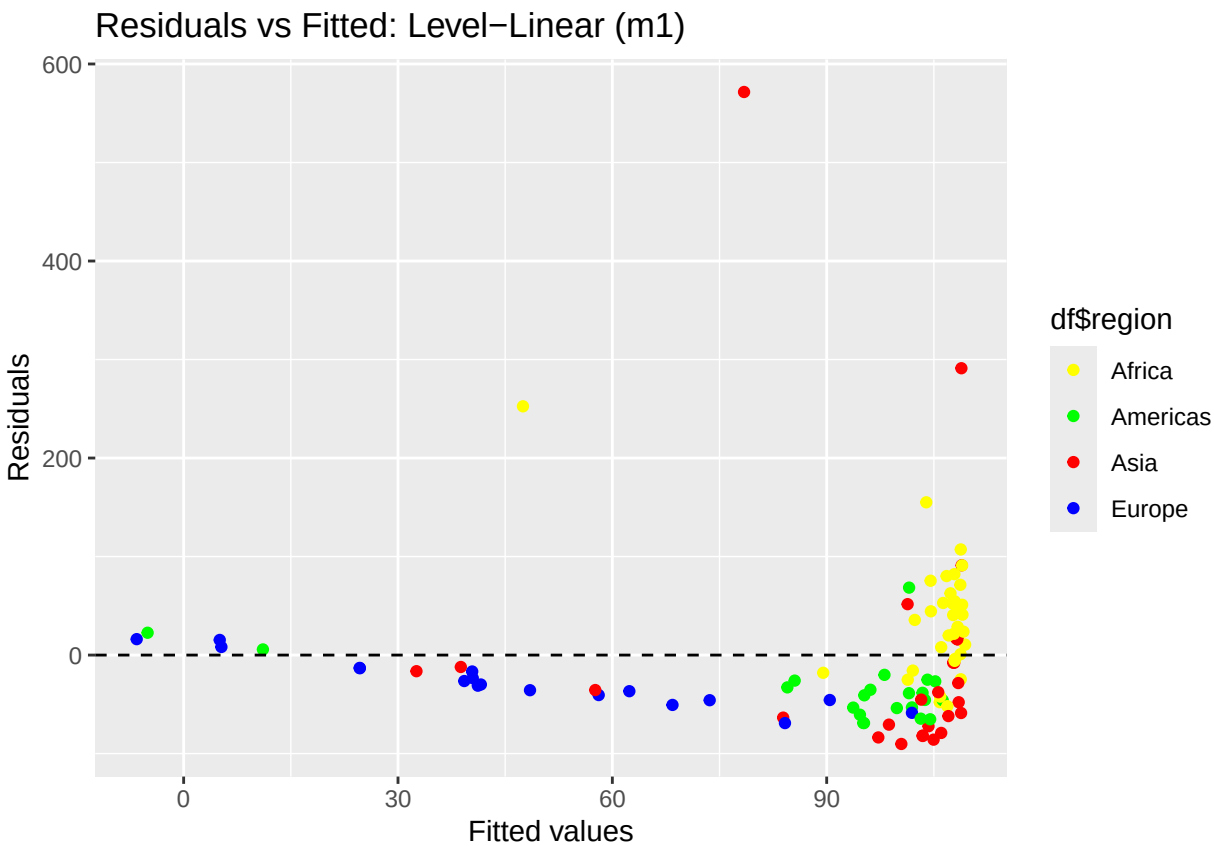
- In m1: what is the predicted change in infant mortality for a $1,000 increase in income?

For a $1,000 increase in income for a country, the model predicts a decrease of 0.0209 in infant mortality rate (20.9 less deaths per 1,000 live births)

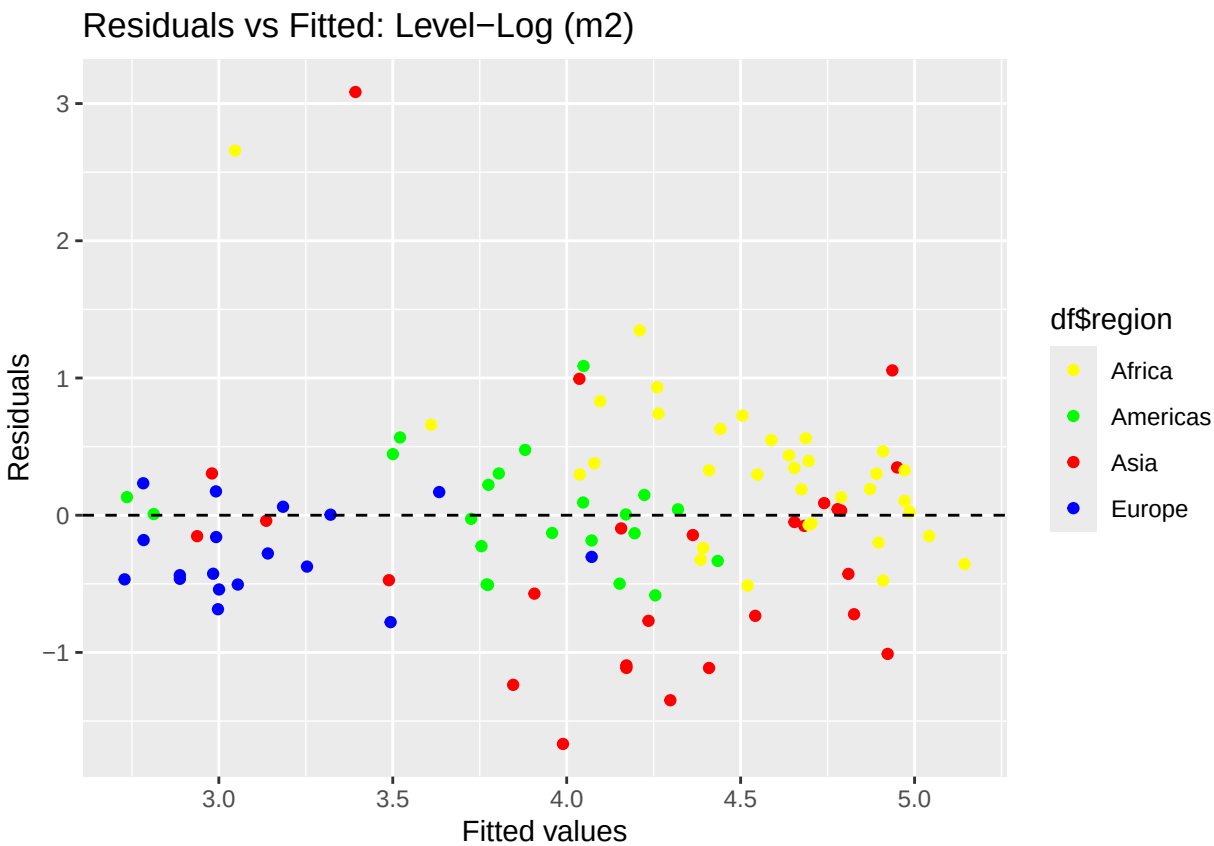
- In m2: recall that the log-log coefficient is an elasticity. What does it mean here? (e.g.,"A 10% increase in income is associated with a % change in infant mortality.")

A 10% increase in income is associated with a -.512% change in infant mortality

```r
m1_aug = augment(m1)
ggplot(m1_aug, aes(x = .fitted, y = .resid, color = df$region)) +
  geom_point() +
  scale_color_manual(values = c("Asia" = "red", "Europe" = "blue",
                                "Americas" = "green", "Africa" = "yellow"))+
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted values", y = "Residuals", title = "Residuals vs Fitted: Level-Linear (m1)")
```




```
m2_aug = augment(m2)
ggplot(m2_aug, aes(x = .fitted, y = .resid, color = df$region)) +
  geom_point() +
  scale_color_manual(values = c("Asia" = "red", "Europe" = "blue",
                                "Americas" = "green", "Africa" = "yellow"))+
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted values", y = "Residuals", title = "Residuals vs Fitted: Level-Log (m2)")
```



The log function has a significantly better residual pattern. The linear model is clustered almost entirely at the bottom right part of the graph with multiple large outliers whereas the log function model is spread more across the line with only a couple outliers.

```
m3 = lm(log(infant) ~ log(income) + region + oil, data = df)
broom::tidy(m3)
```

```
## # A tibble: 6 x 5
##   term           estimate std.error statistic  p.value
##   <chr>             <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       6.55     0.350      18.7  1.14e-33
## 2 log(income)      -0.340    0.0666     -5.10 1.70e- 6
## 3 regionAmericas   -0.550    0.184      -2.98 3.66e- 3
## 4 regionAsia       -0.713    0.158      -4.52 1.75e- 5
## 5 regionEurope     -1.03     0.257      -4.03 1.14e- 4
## 6 oilyes            0.640    0.225       2.84 5.44e- 3
```

The income effect only changes it slightly, with an increase of less than 0.2.

Africa is the reference category in this case, therefore, it is higher than all other regions.

```
avg_slopes(m3)
```

```
##
##    Term          Contrast Estimate Std. Error     z Pr(>|z|)    S   2.5 %
##  income dY/dX             -0.00159   0.000312 -5.10  < 0.001 21.5 -0.0022
##  oil    yes - no           0.64021   0.225052  2.84  0.00444  7.8  0.1991
##  region Americas - Africa -0.54984   0.184492 -2.98  0.00288  8.4 -0.9114
##  region Asia - Africa     -0.71292   0.157572 -4.52  < 0.001 17.3 -1.0218
##  region Europe - Africa   -1.03383   0.256717 -4.03  < 0.001 14.1 -1.5370
```

```
##    97.5 %
##  -0.00098
##   1.08130
##  -0.18824
##  -0.40409
##  -0.53067
##
## Type: response
```

```r
exp(-0.00159)
```

```
## [1] 0.9984113
```

This shows the marginal effects of income, so for every $1 increase in the log of income, there is a decrease by 0.00159 in the log of the infant mortality rate in that country.

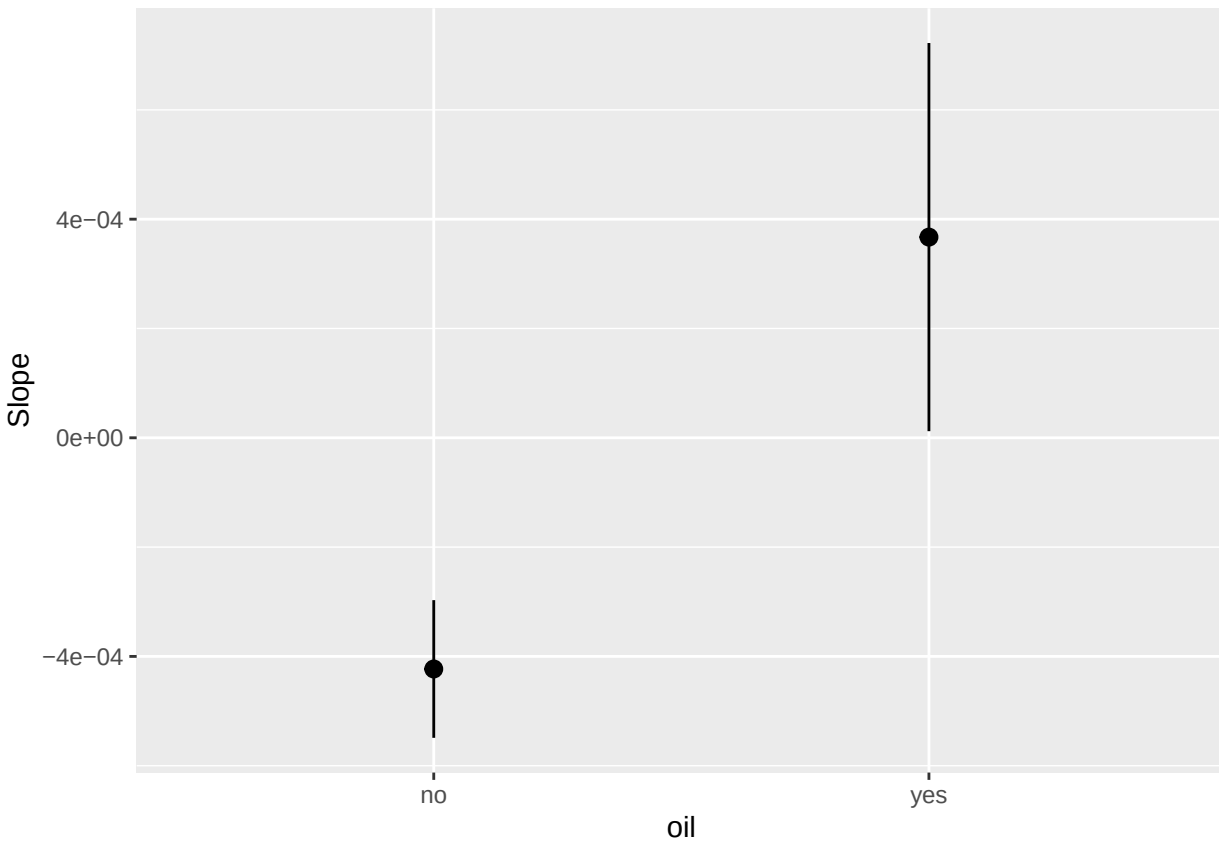
```r
m4 = lm(log(infant) ~ log(income) * oil + region, data = df)
summary(m4)
```

```
##
## Call:
## lm(formula = log(infant) ~ log(income) * oil + region, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.38653 -0.31146 -0.00016  0.31029  1.73208
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         7.00462    0.34216  20.472  < 2e-16 ***
## log(income)        -0.43234    0.06566  -6.584 2.60e-09 ***
## oilyes             -4.38258    1.24379  -3.524 0.000659 ***
## regionAmericas     -0.41618    0.17394  -2.393 0.018714 *
## regionAsia         -0.62925    0.14734  -4.271 4.66e-05 ***
## regionEurope       -0.75910    0.24701  -3.073 0.002772 **
## log(income):oilyes  0.80764    0.19717   4.096 8.89e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5471 on 94 degrees of freedom
## Multiple R-squared:    0.7,  Adjusted R-squared:  0.6808
## F-statistic: 36.55 on 6 and 94 DF,  p-value: < 2.2e-16
```

```r
avg_slopes(m4, variables = "income", by = "oil")
```

```
##
##  oil Estimate Std. Error     z Pr(>|z|)    S     2.5 %   97.5 %
##  no  -0.00209   0.000317 -6.58   <0.001 34.3 -2.71e-03 -0.00147
##  yes  0.00111   0.000549  2.03   0.0427  4.6  3.68e-05  0.00219
##
## Term: income
## Type: response
## Comparison: dY/dX
```

```r
p4 <- plot_slopes(m4, variables = "income", condition = "oil")
p4
```




```
ggsave("slope_plot_Assignment4.2.png", p4, width = 6, height = 4)
```

If the country does not export oil, then for every 1$ increase in per-capita income, there is a decrease of 0.00209 in infant mortality rate. If the country does export oil, then for every 1$ increase in per-capita income, there is an increase of 0.00111 in infant mortality rate. This could be explained by confounding factors such as how developed the country is, whether they are democratic or not, the effectiveness of their healthcare system (or lack thereof). If these factors differ from countries that export oil versus the countries that do not, they could be explanatory for the increase in infant mortality rate when income per capita increases in oil-exporting countries.

```
p1 = predictions(m3,
  newdata = datagrid(
    income = c(1000),
    region = c("Africa"),
    oil = c("no")))
p1
```

```
##
##  income region oil Estimate Std. Error    z Pr(>|z|)     S 2.5 % 97.5 %
##    1000 Africa  no      4.2      0.163 25.8   <0.001 486.7  3.89   4.52
##
## Type: response
```

```
exp(coef(p1))
```

```
##       b1
## 66.98932
```

```r
p2 = predictions(m3,
  newdata = datagrid(
    income = c(20000),
    region = c("Europe"),
    oil = c("no")))
p2
```

```
##
##  income region oil Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
##   20000 Europe  no     2.15      0.195 11.1   <0.001 92.1  1.77   2.53
##
## Type: response
```

```r
exp(coef(p2))
```

```
##       b1
## 8.607374
```

```r
p3 = predictions(m3,
  newdata = datagrid(
    income = c(10000),
    region = c("Americas"),
    oil = c("yes")))
p3
```

```
##
##  income   region oil Estimate Std. Error    z Pr(>|z|)     S 2.5 % 97.5 %
##   10000 Americas yes     3.51      0.276 12.7   <0.001 121.1  2.97   4.05
##
## Type: response
```

```r
exp(coef(p3))
```

```
##      b1
## 33.5281
```

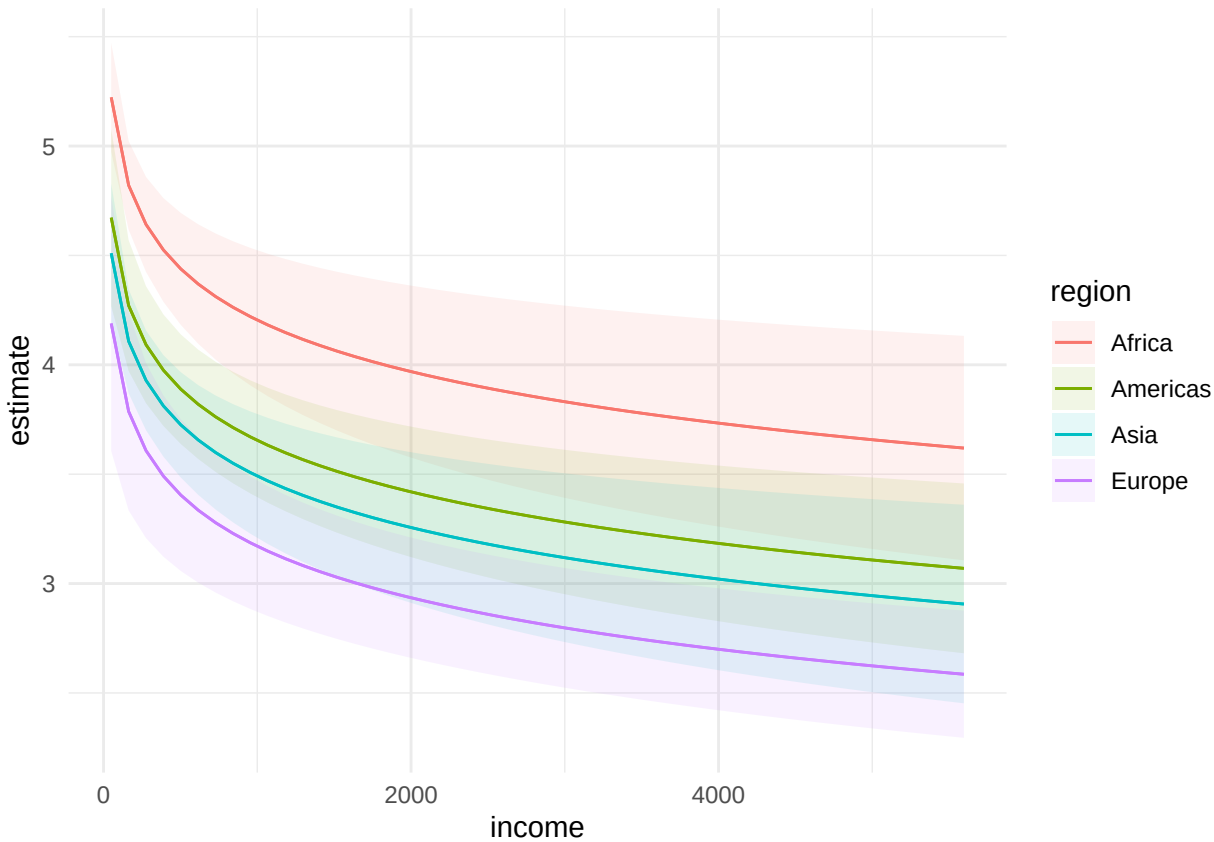
A non-oil African country with an income of $1,000's infant mortality rate is about 67.

A non-oil European country with an income of $20,000's infant mortality rate is about 8.61.

An oil-exporting country in the Americas with an income of $10,000's infant mortality rate is 33.53.

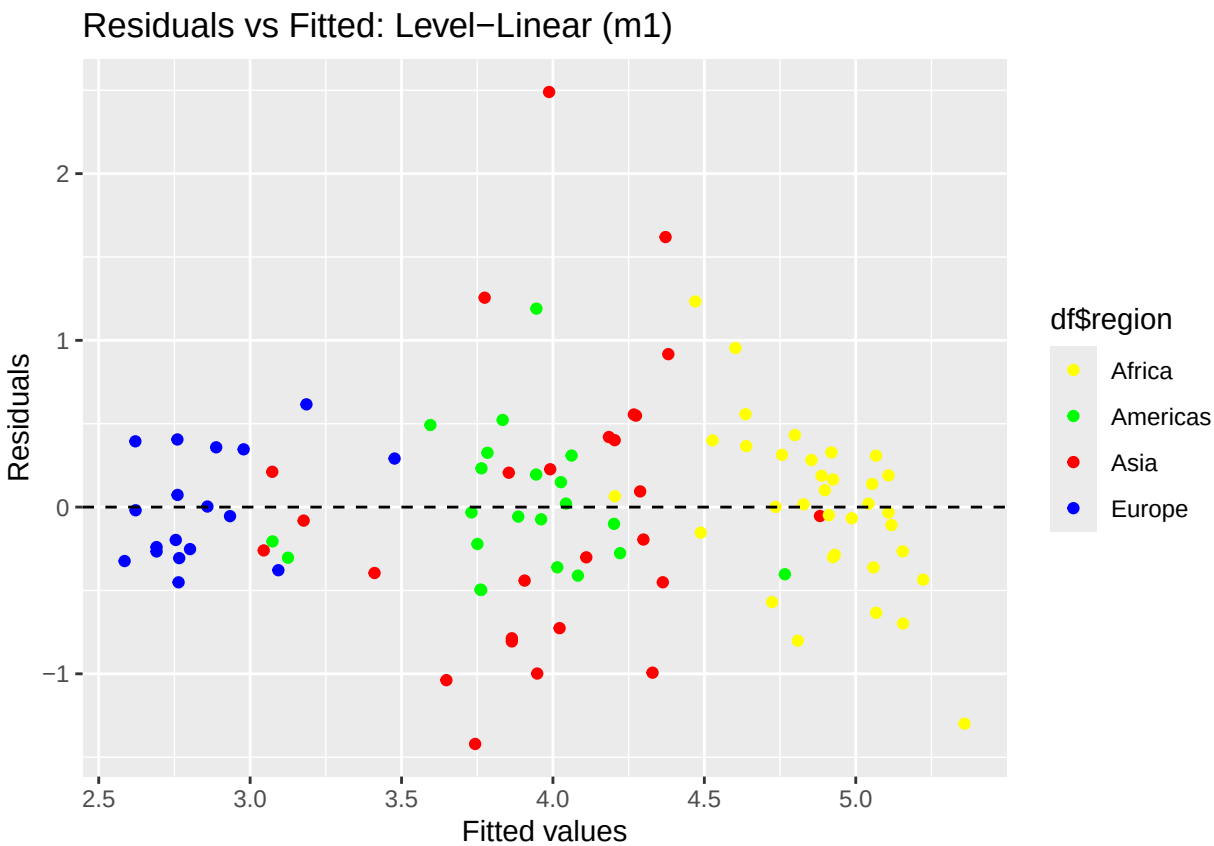
There is a large gap in between the predicted African country and the predicted European country. However, these numbers generally make sense as Africa already had a higher infant mortality rate, especially when you count that it is low-income. Similarly, it is likely that the European country with high income would

```r
plot1 <- plot_predictions(m3, condition = list("income", "region"))+
  theme_minimal()
plot1
```



This predicted plot shows that as the income level of a country increases, infant mortality rate decreases. The decline in infant mortality rate is steep as income first begins to increase, then slows as income becomes greater than $1,000. Europe overall has the lowest infant child mortality rate, followed by Asia, with the Americas slightly higher. Africa is predicted to have the overall highest infant mortality rate, no matter the income level. The main limitations of this study is that there may be confounding factors affecting infant mortality rate, not whether it is an oil-exporting country or not. Similarly, how full and accurate is the data? How was the data collected? We cannot be positive that the infant mortality rate is entirely accurate, especially in countries where healthcare is less formalized, possibly underreporting (or overreporting) infant deaths.

```
m3_aug = augment(m3)
ggplot(m3_aug, aes(x = .fitted, y = .resid, color = df$region)) +
  geom_point() +
  scale_color_manual(values = c("Asia" = "red", "Europe" = "blue",
                                "Americas" = "green", "Africa" = "yellow"))+
  geom_hline(yintercept = 0, linetype = "dashed")+
  labs(x = "Fitted values", y = "Residuals", title = "Residuals vs Fitted: Level-Linear (m1)")
```




```r
modelsummary(
  list("Level" = m1, "Log-Log" = m2,
       "Controls" = m3, "Interaction" = m4),
  vcov = "robust",
  stars = TRUE,
  gof_map = c("r.squared", "nobs"))
```


```r
modelsummary(
  list("Level" = m1, "Log-Log" = m2,
       "Controls" = m3, "Interaction" = m4),
  stars = TRUE,
  gof_map = c("r.squared", "nobs"))
```


These results do not demonstrate heteroskedasticity as the standard errors are not increasing or decreasing due to the increase in the dependent variable.

In this instance, the standard errors do not change when using robust versus default standard errors. However, we should still use it as there is no downside to using it and it will always ensure homoskedasticity, whether the default standard errors are heteroskedastic or not.

	Level	Log-Log	Controls	Interaction
(Intercept)	110.421***	7.146***	6.552***	7.005***
	(9.037)	(0.276)	(0.398)	(0.282)
income	−0.021***			
	(0.004)			
log(income)		−0.512***	−0.340***	−0.432***
		(0.049)	(0.078)	(0.057)
regionAmericas			−0.550**	−0.416*
			(0.184)	(0.162)
regionAsia			−0.713***	−0.629**
			(0.207)	(0.203)
regionEurope			−1.034***	−0.759***
			(0.256)	(0.211)
oilyes			0.640	−4.383
			(0.472)	(3.019)
log(income) × oilyes				0.808+
				(0.481)
R2	0.109	0.502	0.646	0.700
Num.Obs.	101	101	101	101

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

	Level	Log-Log	Controls	Interaction
(Intercept)	110.421***	7.146***	6.552***	7.005***
	(10.539)	(0.317)	(0.350)	(0.342)
income	−0.021***			
	(0.006)			
log(income)		−0.512***	−0.340***	−0.432***
		(0.051)	(0.067)	(0.066)
regionAmericas			−0.550**	−0.416*
			(0.184)	(0.174)
regionAsia			−0.713***	−0.629***
			(0.158)	(0.147)
regionEurope			−1.034***	−0.759**
			(0.257)	(0.247)
oilyes			0.640**	−4.383***
			(0.225)	(1.244)
log(income) × oilyes				0.808***
				(0.197)
R2	0.109	0.502	0.646	0.700
Num.Obs.	101	101	101	101

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001